Multilevel partitioning

A multilevel V-cycle coarsens the graph, partitions the tiny problem, projects the assignment back

The idea

- Coarsening shrinks the search space
- The coarse partition is cheap
- Projection gives a feasible fine assignment that FM or KL can polish
- A bug in coarsening poisons every finer level

Pseudocode

- Multilevel is a V-cycle in pseudocode
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- On TINY_GRAPH the projected seed is already ABC versus DE at cut three
- Refine keeps that cut, multilevel beats polishing the cut-twelve bad seed alone

Algorithm sketch — try these

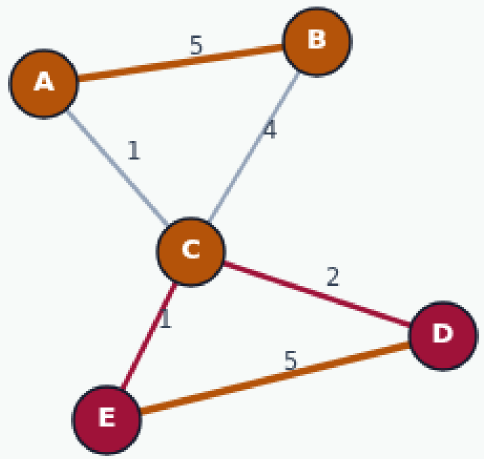
```
INPUT: G, coarsen until tiny
OUTPUT: fine side[]
coarsen: match/cluster heavy edges
partition coarse (spectral/KL/FM)
project labels to finer level
refine with FM/KL at each uncoarsen
GOLDEN project ABC|DE cut=3; refine keeps 3
```

Coarsen with greedy merge

MULTILEVEL PARTITION

Coarsen with greedy merge

Step 1 / 5 · coarsen · module04-01-multilevel-partition



Explanation

Multilevel V-cycle contracts the graph via greedy pair merge to coarseK=2. Supernodes C1={A,B,C} and C2={D,E} capture community structure.

- Greedy heaviest-edge merges
- Log each contraction
- Coarse graph has 2 clusters

```
coarseK=2
labels: C1, C2
```

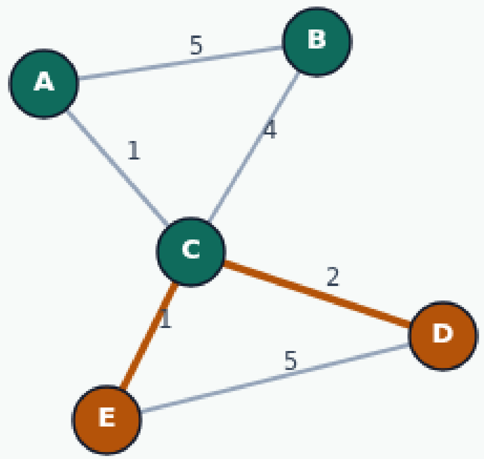
Coarsen with greedy merge

Project to fine bipartition

MULTILEVEL PARTITION

Project to fine bipartition

Step 2 / 5 · project · module04-01-multilevel-partition



Explanation

Map coarse clusters to sides 0/1: ABC → 0, DE → 1. Projected seed ABC|DE already has cutsize 3 — unlike BAD_SEED's cut of 12.

- Projection preserves membership
- Good global seed from coarsening
- Ready for local FM refine

```
project: ABC|DE
cutsize: 3
BAD_SEED cut: 12
```

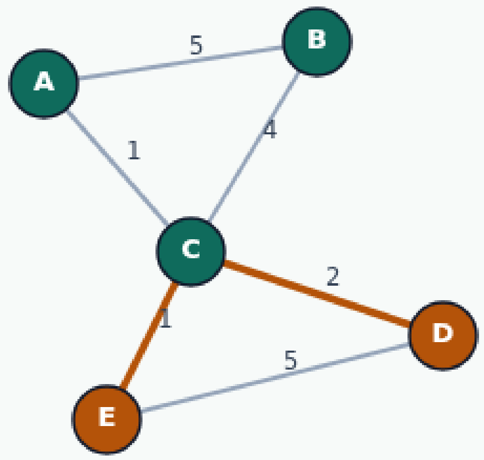
Project to fine bipartition

FM refine on fine graph

MULTILEVEL PARTITION

FM refine on fine graph

Step 3 / 5 · refine · module04-01-multilevel-partition



Explanation

FM polishes the projected seed. On TINY_GRAPH the projection is already optimal — refine keeps ABC|DE at cutsize 3.

- Single-vertex moves with rollback
- Local search on fine graph
- No improvement needed here

```
refine: ABC|DE
cutsize: 3
```

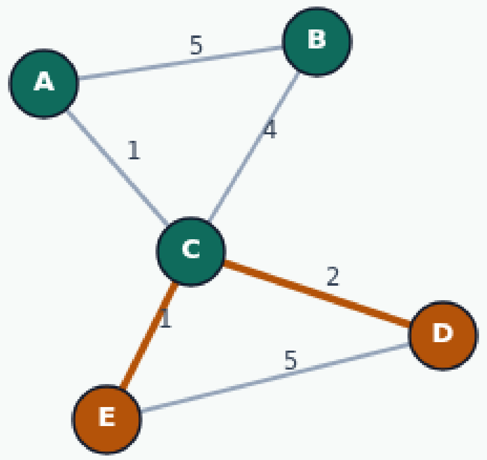
FM refine on fine graph

Final P0/P1 labels

MULTILEVEL PARTITION

Final P0/P1 labels

Step 4 / 5 · final · module04-01-multilevel-partition



Explanation

Output renames sides to P0/P1 for placers. Final communities ABC|DE with cutsize 3 — multilevel beats refining a random bad seed alone.

- P0 = ABC, P1 = DE
- cutsize: 3
- V-cycle complete

```
final: ABC|DE
cutsize: 3
labels: P0, P1
```

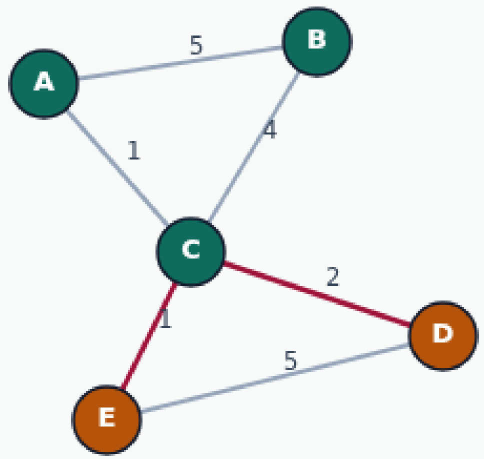
Final P0/P1 labels

Multilevel V-cycle mindset

MULTILEVEL PARTITION

Multilevel V-cycle mindset

Step 5 / 5 · takeaway · module04-01-multilevel-partition



Explanation

Coarsen for global structure, project, refine locally. Real placers nest this in repeated V-cycles with hyperedges and balance — this lab is the skeleton.

- coarsen → project → refine
- Beats BAD_SEED cut 12 → 3
- Industry default for large netlists

```
Starter golden: cut=3
```

Multilevel V-cycle mindset

Browser lab track

- In the browser lab track, open the **multilevel-partition** lab from the tools shelf
- Load the starter graph, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse the tiny graph, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps
- For multilevel flows, verify coarsening before you blame the refiner

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you’re ready, take the short quiz, then continue to the next module